ESSENTIALS OF WEB PROGRAMMING
UNIT-I

1. INTRODUCTION TO WEB AND INTERNET

1.1 INTRODUCTION

The last three decades have witnessed tremendous growth in information and communication technologies. Computers are no longer isolated machines used only for local computations. Today, computers communicate with each other, exchange information, share resources, and provide services across the world. This global communication is made possible through the Internet.

The Internet has transformed almost every aspect of human life, including education, healthcare, banking, entertainment, business, and scientific research. Applications such as Google, YouTube, Amazon, WhatsApp, and ChatGPT operate through the Internet.
The World Wide Web (WWW) is one of the most important services provided by the Internet. The Web enables users to access information in the form of web pages and websites using web browsers.

Therefore, before learning HTML and web programming, it is essential to understand the concepts of the Internet and the World Wide Web.

1.2 LEARNING OBJECTIVES

1. Understand the concept of the Internet.
2. Define the World Wide Web.
3. Explain the history of the Internet.
4. Differentiate between Internet and WWW.
5. Describe web pages and websites.
6. Explain the role of browsers and web servers.
7. Understand the working of the Web.

1.3 NEED FOR THE INTERNET

Before the Internet, information sharing was difficult because:

- Communication was slow.
- Documents had to be physically transported.
- Access to information was limited.
- Collaboration among researchers was difficult.
- Geographical barriers restricted communication.

The Internet solved these problems by providing:

- Fast communication.
- Global connectivity.

- Information sharing.
- Resource sharing.
- Online services.

Example

- A student in India can attend an online lecture conducted in the United States.
- A doctor in one city can examine medical reports sent from another city.
- A customer can purchase products from another country.
- All these activities are possible because of the Internet.

1.4 HISTORY OF THE INTERNET

The origin of the Internet can be traced back to the late 1960s.The United States Department of Defense developed a communication network known as ARPANET (Advanced Research Projects Agency Network). The objective of ARPANET was to establish a communication system that could continue functioning even if some communication links failed. Later, universities and research organizations joined these networks. This development led to the creation of the modern Internet.

EVOLUTION OF INTERNET

ARPANET (1969)
↓
NSFNET (1980s)
↓
Academic Networks
↓
Commercial Networks
↓
INTERNET

1.5 INTERNET

Definition: The Internet is a global system of interconnected computer networks that communicate using standard communication protocols.

Alternative Definition: The Internet is a network of networks that connects millions of computers and electronic devices throughout the world.

CHARACTERISTICS OF INTERNET

The important characteristics of the Internet are:

1. Global connectivity.
2. Resource sharing.
3. High-speed communication.

4. Scalability.
5. Reliability.
6. Decentralization.

COMPONENTS OF INTERNET

The Internet consists of:

1. **Client Computers:** Devices used by users.
   Examples: Mobile phones, Laptops, Desktops

2. **Servers:** Computers that provide services and information.
   Examples: Google servers, Amazon servers

3. **Communication Media:** Optical fiber cables, Wireless communication, Satellite links

4. **Routers:** Devices that transfer data between networks.

5. **Protocols:** Rules governing communication.
   Examples: TCP/IP, HTTP

1.6 WORLD WIDE WEB (WWW)

Definition: The World Wide Web is a collection of interconnected web pages and documents that are accessed through the Internet using web browsers.

Alternative Definition: WWW is an information system consisting of websites and web pages linked together by hyperlinks.

The Web was proposed by Sir Tim Berners-Lee in 1989 while working at CERN.
His objective was to enable researchers to share information efficiently.

TECHNOLOGIES INTRODUCED

The Web introduced three major technologies:

- HTML
- HTTP
- URL

These technologies became the foundation of modern web systems.

Figure 1.1 STRUCTURE OF WWW



↓

Hyperlinks

1.7 INTERNET Vs WORLD WIDE WEB

Although these terms are often used interchangeably, they are different.

Internet	World Wide Web
Global network	Information system
Provides connectivity	Provides information
Uses TCP/IP	Uses HTTP
Infrastructure	Service
Hardware and software	Web pages and websites

Observation:

- The Internet can exist without the Web.
- However, the Web cannot exist without the Internet.

1.8 WEB PAGE

Definition: A web page is a document created using HTML and displayed by a web browser.

Examples: Google home page, Amazon product page, University result page.

Example: HTML Page

```
<!DOCTYPE html>
<html>

<head>
    <title>My First Page</title>
</head>

<body>
    <h1>Welcome Students</h1>

    <p>This is my first webpage.</p>
</body>

</html>
```

Explanation: The browser interprets HTML instructions and displays the contents to the user.

1.9 WEBSITE

Definition: A website is a collection of related web pages connected through hyperlinks.

Examples: Amazon, Facebook, University website.

A university website may contain:

- Home page
- Admissions page
- Results page
- Contact page

Together, these pages form a website.

1.10 WEB BROWSER

Definition: A web browser is a software application used to access and display web pages.

Examples: Google Chrome, Mozilla Firefox, Microsoft Edge, Safari.

FUNCTIONS OF BROWSER

- Sends requests.
- Receives responses.
- Displays webpages.
- Executes HTML instructions.

1.11 WEB SERVER

Definition: A web server is a computer that stores web pages and delivers them to users upon request.

Examples: Apache, Nginx, IIS

1.12 URL (UNIFORM RESOURCE LOCATOR)

Definition: A URL is the address of a web page on the Internet.

Example: https://www.google.com

Components of URL: https://www.google.com

https → Protocol

www → Subdomain

google → Domain Name

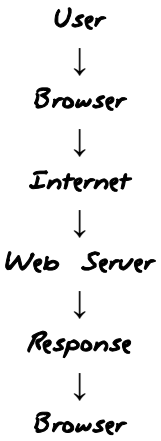
.com → Domain Extension

1.13 WORKING OF THE WEB

When a user enters a URL, the following sequence occurs:

- The user enters the URL.
- The browser sends a request.
- The Internet transfers the request.
- The web server processes the request.
- The server sends a response.
- The browser displays the webpage.

Figure 1.2 WORKING OF THE WEB



1.14 APPLICATIONS OF INTERNET AND WEB

The Internet and Web have applications in:

Education

- Online classes
- Learning portals

Banking

- Internet banking
- UPI services

Healthcare

- Online consultation

- Appointment systems

E-Commerce

- Amazon
- Flipkart

Entertainment

- YouTube
- Netflix

Artificial Intelligence

- ChatGPT
- Gemini

1.15 ADVANTAGES

- Fast communication.
- Global connectivity.
- Information sharing.
- Resource sharing.
- Online services.
- Remote education.

1.16 LIMITATIONS

- Security threats.
- Privacy issues.
- Cybercrime.
- Fake information.
- Internet addiction.

IMPORTANT POINTS

✓ Internet and WWW are different.
✓ WWW was proposed by Tim Berners-Lee.
✓ HTML, HTTP, and URL are the foundations of the Web.
✓ Browsers display web pages.
✓ Servers store web pages.

Question

1. Can the Internet exist without Google?
Answer: Yes.

2. Can Google exist without the Internet?

Answer: No.

SCENARIO-BASED EXAMPLES

1. A hospital sends patient reports to another branch, which technology enables communication?

   Answer: Internet.

2. A student accesses multiple university pages, which system provides these pages?

   Answer: WWW.

3. An organization stores webpages on a powerful computer, what is that computer called?

   Answer: Web Server.

4. A user enters: www.amazon.com,what identifies this address?

   Answer: URL.

SUMMARY

- The Internet is a global communication network.
- The World Wide Web is an information system built on the Internet.
- Browsers access web pages stored on web servers.
- URLs identify web resources.
- The Web has revolutionized communication, education, business, and entertainment.

The Internet connects computers, whereas the World Wide Web connects information and people through interconnected web pages.

2. EVOLUTION OF HTML

2.1 INTRODUCTION

The World Wide Web has undergone tremendous changes since its inception. In the early days, web pages contained only simple text and hyperlinks. Modern web applications, however, support multimedia, graphics, animations, mobile devices, and interactive content. The language responsible for this evolution is **HTML (HyperText Markup Language)**.

HTML has evolved through several versions to meet the increasing demands of web applications. Each version introduced new features and improved the limitations of previous versions. Understanding the evolution of HTML helps students appreciate the development of modern web technologies and understand the significance of HTML5.

2.2 LEARNING OBJECTIVES

After studying this topic, the learner will be able to:

- Explain the need for HTML.
- Discuss the evolution of HTML.
- Describe different HTML versions.
- Compare older HTML versions with HTML5.
- Understand the importance of HTML5.
- Explain the major improvements introduced in HTML.

2.3 NEED FOR HTML

Before HTML, sharing information between computers was difficult because:

- Different computers used different document formats.
- Documents were not interconnected.
- Information retrieval was difficult.
- Scientific researchers could not efficiently share data.

Tim Berners-Lee proposed HTML to solve these problems.
The objective was: To create a common language for sharing information through interconnected documents.

Example: Researchers in different countries wanted to access the same scientific documents. HTML made this possible by creating linked electronic documents.

2.4 WHAT IS EVOLUTION OF HTML?

Definition: The evolution of HTML refers to the gradual development of HTML from a simple document formatting language to a powerful web development language.

Over the years, HTML introduced:

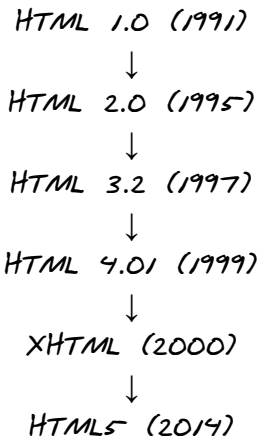
- Images

- Tables
- Forms
- CSS support
- Multimedia
- Semantic elements
- Mobile compatibility

2.5 HISTORY OF HTML

In 1989, Tim Berners-Lee, working at CERN, proposed a system for sharing scientific documents. In 1991, the first version of HTML was introduced. Since then, HTML has evolved continuously.

Figure 2.1 Evolution of HTML



2.6 HTML 1.0 (1991)

HTML 1.0 was the first version of HTML.

Features:

- Headings
- Paragraphs
- Hyperlinks
- Basic formatting

Limitations

- No images.
- No tables.
- No forms.
- No multimedia.

Example

Program 2.1 Basic HTML 1.0 Page

```
<html>
```

```
<body>

<h1>Welcome</h1>

<p>Hello Students</p>

</body>

</html>
```

Observation: HTML 1.0 was suitable only for static text documents.

2.7 HTML 2.0 (1995)

HTML 2.0 standardized HTML specifications.
Major Feature:
- Forms

This allowed websites to collect user information.

Example Applications
- Registration forms
- Login pages
- Feedback forms

Example

```
<form>

<input type="text">

<input type="submit">

</form>
```

2.8 HTML 3.2 (1997)

HTML 3.2 introduced:
- Tables
- Images
- Fonts
- Applets

Example Applications
- News websites
- Product catalogs
- Educational websites

Example

```
<table border="1">

<tr>
<td>Student</td>
</tr>

</table>
```

2.9 HTML 4.01 (1999)

HTML 4.01 focused on separating content and presentation.
Major improvements:

- CSS support
- Scripting support
- Improved forms
- Better accessibility

Observation

HTML became more structured and maintainable.

2.10 XHTML (2000)

XHTML stands for: Extensible HyperText Markup Language, XHTML introduced stricter coding rules.

Rules:

- All tags must be closed.
- Tags must be lowercase.
- Proper nesting is mandatory.

Incorrect XHTML

```
<p>Hello
```

Correct XHTML

```
<p>Hello</p>
```

Limitation

Developers found XHTML too restrictive.

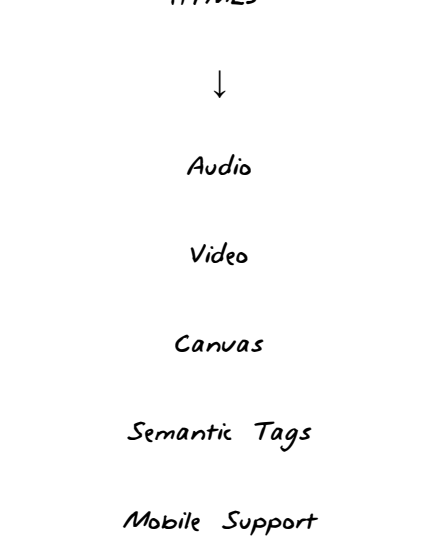
2.11 HTML5 (2014)

HTML5 is the latest major version of HTML.

HTML5 introduced:

- Audio support
- Video support
- Semantic elements
- Canvas
- Local storage
- Mobile compatibility

Figure 2.2 Features of HTML5



Example

Program 2.2 Video in HTML5

```
<video controls>
<source src="lecture.mp4">
</video>
```

Example

Program 2.3 Audio in HTML5

```
<audio controls>
<source src="song.mp3">
</audio>
```

2.12 COMPARISON OF HTML VERSIONS

Version	Major Feature
HTML 1.0	Basic pages
HTML 2.0	Forms
HTML 3.2	Tables and images
HTML 4.01	CSS support
XHTML	Strict rules
HTML5	Multimedia

2.13 ADVANTAGES OF HTML5

- Supports multimedia.
- Mobile friendly.
- Better semantics.
- Improved accessibility.
- Reduced dependency on plugins.
- Faster performance.

2.14 APPLICATIONS OF HTML5

HTML5 is used in:

- YouTube
- Netflix
- Online education
- Web applications
- Mobile applications
- Interactive websites

Example

YouTube previously required Flash.
Today:
<video>
plays videos directly.

SCENARIO-BASED EXAMPLES

1: A company wants to create online registration forms, Which HTML version introduced forms?
Answer: HTML 2.0

2: A shopping website wants product tables, which version introduced tables?
Answer: HTML 3.2

3: A website wants to play videos without Flash, which version should be used?

Answer: HTML5

4: A developer wants all tags to be properly closed.
Which standard enforces this?
Answer: XHTML

5: A mobile-friendly website requires multimedia support, Which version is suitable?
Answer: HTML5

IMPORTANT POINTS

✓ HTML was developed by Tim Berners-Lee.
✓ HTML 1.0 introduced basic web pages.
✓ HTML 2.0 introduced forms.
✓ HTML 3.2 introduced tables.
✓ HTML 4.01 supported CSS.
✓ XHTML introduced strict rules.
✓ HTML5 introduced multimedia.

Wrong: HTML5 is a programming language.
Correct: HTML5 is a markup language.

Wrong: XHTML replaced HTML.
Correct: HTML5 became more popular than XHTML.

SUMMARY

- HTML was introduced in 1991.
- Several versions improved HTML capabilities.
- HTML evolved from simple text pages to multimedia web applications.
- HTML5 is the latest major version.
- Modern websites heavily depend on HTML5.

The evolution of HTML reflects the evolution of the Web itself—from static documents to modern interactive web applications.

3. WORLD WIDE WEB (WWW)

3.1 INTRODUCTION

The World Wide Web (WWW) is one of the most important services available on the Internet. Today, millions of people use the Web to access information, communicate, perform online transactions, study, watch videos, and use various online services. Although the terms **Internet** and **World Wide Web** are often used interchangeably, they are different. The Internet provides the communication infrastructure, whereas the World Wide Web provides access to information through web pages and websites.

The Web has transformed the world into a global information system where users can access information from anywhere at any time.

3.2 LEARNING OBJECTIVES

- Define World Wide Web.
- Explain the need for WWW.
- Discuss the history of WWW.
- Describe the components of WWW.
- Explain the working of WWW.
- Differentiate WWW from the Internet.
- Understand the applications of WWW.

3.3 NEED FOR WORLD WIDE WEB

Before the Web was introduced:

- Documents were stored separately.
- Information sharing was difficult.
- Researchers could not easily access documents.
- There was no common information system.
- Finding information was time-consuming.

The World Wide Web solved these problems by providing:

- Interconnected documents.
- Hyperlinks.
- Global accessibility.
- Easy navigation.
- Information sharing.

Example

- A researcher in India can access a scientific paper stored in Europe.
- A student can access lecture notes stored on a university server.
- These facilities became possible because of the World Wide Web.

3.4 DEFINITION OF WORLD WIDE WEB

Definition: The World Wide Web (WWW) is a system of interconnected web pages and documents that are accessed through the Internet using web browsers.

Definition: WWW is a collection of websites and web pages connected through hyperlinks.

Definition: The World Wide Web is an information retrieval system in which documents and resources are identified by URLs and accessed through the Internet.

3.5 HISTORY OF WWW

The World Wide Web was invented by: **Tim Berners-Lee**
Year: **1989**
Place: **CERN, Switzerland**
The primary objective was: To share scientific information among researchers.

Tim Berners-Lee introduced:

- HTML
- HTTP
- URL

These technologies became the foundation of the World Wide Web.

Figure 3.1 Development of WWW



3.6 COMPONENTS OF WWW

The World Wide Web consists of the following components:

1. Web Pages
2. Websites
3. Web Browsers
4. Web Servers
5. URLs
6. Hyperlinks
7. HTTP Protocol

1. Web Pages

A web page is an HTML document displayed by a browser.

Examples:

- Home page
- Contact page
- Results page

2. Websites

A website is a collection of related web pages.

Examples:

- Google
- Amazon
- University websites

3. Web Browser

A browser is software used to access web pages.

Examples:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

4. Web Server

A web server stores websites and delivers them to users.

Examples:

- Apache
- Nginx
- IIS

5. URL

URL identifies the location of a web page.

Example: https://www.google.com

6. Hyperlink

Hyperlinks connect one web page to another.

Example: "Click Here"

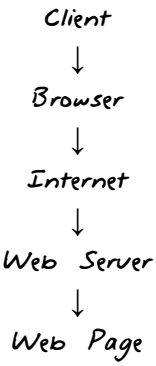
7. HTTP

HTTP stands for:
HyperText Transfer Protocol.
It transfers web pages between browsers and servers.

3.7 ARCHITECTURE OF WWW

- The Web follows the Client-Server architecture.
- The user acts as the client.
- The server provides information.

Figure 3.2 WWW Architecture



3.8 WORKING OF WWW

The working of the World Wide Web involves the following steps:

Step 1: The user enters a URL.

Step 2: The browser sends a request.

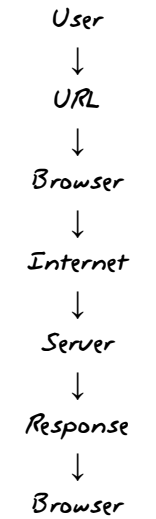
Step 3: The request travels through the Internet.

Step 4: The server receives the request.

Step 5: The server sends the web page.

Step 6: The browser displays the page.

Figure 3.3 Working of WWW



Example

Suppose a student enters:

https://www.google.com

The browser requests Google's server.

Google sends the webpage.
The browser displays the page.

3.9 CHARACTERISTICS OF WWW

The important characteristics are:

- Hypertext-based.
- Global accessibility.
- Interconnected documents.
- Multimedia support.
- User-friendly navigation.
- Platform independent.

3.10 ADVANTAGES OF WWW

- Easy access to information.
- Fast communication.
- Global connectivity.
- Multimedia support.
- Educational resources.
- E-commerce applications.
- Online services.

3.11 LIMITATIONS OF WWW

- Security risks.
- Fake information.
- Cybercrime.
- Privacy issues.
- Dependency on the Internet.

3.12 APPLICATIONS OF WWW

The World Wide Web is widely used in:

Education

- Online classes
- Digital libraries

Banking

- Net banking
- Online transactions

Healthcare

- Online appointments
- Medical records

E-Commerce
- Amazon
- Flipkart

Entertainment
- YouTube
- Netflix

Government Services
- Online certificates
- Tax services

Artificial Intelligence
- ChatGPT
- Gemini

Program 3.1 Simple Web Page

```
<!DOCTYPE html>

<html>
<head>
      <title>WWW Example</title>
</head>
<body>
<h1>World Wide Web</h1>
<p>WWW provides information through web pages.</p>
</body>
</html>
```

Explanation:

- <html> defines the document.
- <head> stores metadata.
- <title> specifies browser title.
- <body> contains visible content.

Observation: The browser interprets HTML pages and displays information to users.

"Can WWW exist without the Internet?"
Answer: No.

"Can the Internet exist without WWW?"
Answer: Yes.

SCENARIO-BASED EXAMPLES

1: A student accesses the university results page.
Which system provides the webpage?
Answer: WWW.

2: A hospital website contains appointment, doctor, and patient pages.
What is this collection called?
Answer: Website.

3: A user types www.amazon.com.
What identifies this address?
Answer: URL.

4: A browser requests information from Google.
Which computer provides the webpage?
Answer: Web Server.

5: A student clicks "View Results."
What enables movement between pages?
Answer: Hyperlink.

IMPORTANT POINTS

✓ WWW was proposed by Tim Berners-Lee.
✓ WWW uses HTML, HTTP, and URLs.
✓ WWW follows client-server architecture.
✓ Browsers access web servers.
✓ Web pages form websites.
✓ WWW operates on the Internet.

SUMMARY

- WWW is a collection of interconnected web pages.
- Tim Berners-Lee invented the Web.
- HTML, HTTP, and URL form the foundation of WWW.
- Browsers communicate with servers.
- WWW provides information through websites.
- Modern web applications depend on the World Wide Web.

The World Wide Web transforms the Internet into a global information system by connecting people through websites and web pages.

4. HTML (HYPERTEXT MARKUP LANGUAGE)

4.1 INTRODUCTION

The World Wide Web contains millions of web pages that provide information, services, entertainment, education, and communication. These web pages are created using a standard language called HTML.

HTML is the fundamental building block of every website. It provides the structure and organization of web pages. Without HTML, web browsers cannot display web content properly.
Every website, whether it is Google, Amazon, YouTube, or a university website, uses HTML as its foundation.

4.2 LEARNING OBJECTIVES

1. Define HTML.
2. Explain the meaning of HyperText.
3. Explain the meaning of Markup.
4. Understand the purpose of HTML.
5. Explain the features of HTML.
6. Describe the structure of an HTML document.
7. Differentiate HTML from programming languages.

4.3 WHAT IS HTML?

Definition: HTML stands for HyperText Markup Language, HTML is a standard markup language used to create and structure web pages.

Definition: HTML is a language used to describe the structure and presentation of information on web pages.

Definition: HTML is a markup language that uses tags and elements to define the structure and content of web documents.

4.4 FULL FORM OF HTML

HTML

↓

H - Hyper
T - Text
M - Markup
L - Language

Each term has a specific meaning.

4.5 HYPERTEXT

Definition: HyperText refers to text that contains links to other documents or web pages. HyperText allows users to move from one page to another by clicking hyperlinks.

Example:

When a user clicks:
"Click Here"
and another page opens, it is called HyperText.
Examples:
- Google search results.
- Wikipedia links.
- University websites.
- Online shopping websites.

4.6 TEXT

Text refers to the information displayed on a web page.
Examples:
- Headings
- Paragraphs
- Articles
- News
- Descriptions

HTML organizes this text for display.

4.7 MARKUP

Definition: Markup means adding instructions or tags to text to describe its appearance and structure.

Example:
The text:
College of Engineering
can be marked as:
The browser understands that this is a heading.
Thus, HTML uses tags to mark content.

4.8 LANGUAGE

HTML has:
- Rules
- Syntax
- Structure
- Standards

Therefore, it is called a language.

However, HTML is not a programming language because it does not perform calculations or logical operations.

4.9 WHY HTML WAS DEVELOPED

In 1989, Tim Berners-Lee developed HTML.

The objectives were:

- Share scientific documents.
- Connect information.
- Create electronic documents.
- Provide a standard document format.

HTML solved the problem of sharing information between different computers.

4.10 FEATURES OF HTML

The important features of HTML are:

1. Easy to learn.
2. Platform independent.
3. Tag based language.
4. Supports hyperlinks.
5. Supports multimedia.
6. Supports forms.
7. Browser independent.
8. Free and open standard.

4.11 CHARACTERISTICS OF HTML

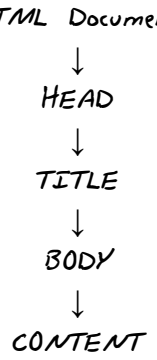
- Simple syntax.
- Case insensitive (HTML5).
- Uses tags.
- Supports nested elements.
- Supports multimedia.
- Integrates with CSS and JavaScript.

4.12 HTML DOCUMENT STRUCTURE

Every HTML document contains the following sections:

1. DOCTYPE Declaration
2. HTML Element
3. HEAD Section
4. TITLE Element
5. BODY Section

Figure 4.1 Structure of HTML Document



Program 4.1 Basic HTML Program

```
<!DOCTYPE html>

<html>

<head>

<title>My First Page</title>

</head>

<body>

<h1>Hello World</h1>

<p>This is my first webpage.</p>

</body>

</html>
```

4.13 WHY HTML IS NOT A PROGRAMMING LANGUAGE

HTML cannot:

- Perform calculations.
- Execute loops.
- Make decisions.
- Create functions.

HTML only describes content.

Example:

HTML:

```
<h1>Welcome</h1>
```

Programming language:

```
if(age > 18)
{
    vote();
}
```

Therefore: HTML is a Markup Language and not a Programming Language.

4.14 ADVANTAGES OF HTML

1. Simple and easy.
2. Free to use.
3. Supported by all browsers.
4. Platform independent.
5. Integrates with CSS.
6. Integrates with JavaScript.
7. Suitable for beginners.

4.15 LIMITATIONS OF HTML

1. Cannot perform calculations.
2. Cannot handle databases.
3. Cannot create dynamic applications alone.
4. Requires CSS for design.
5. Requires JavaScript for interactivity.

4.16 APPLICATIONS OF HTML

HTML is used in:

- Websites.
- Web applications.
- Educational portals.
- E-commerce sites.
- Banking applications.
- Hospital systems.
- Online examinations.
- Government websites.

Examples:

- Google
- Amazon
- YouTube
- University websites

SCENARIO-BASED EXAMPLES

1: A company wants to display information on a browser.
Which language should be used?
Answer: HTML.

2: A webpage contains clickable text that opens another page.
Which term in HTML supports this?
Answer: HyperText.

3: A browser interprets instructions enclosed within angle brackets.
What are these instructions called?
Answer: Tags.

4: A student creates:
What type of language is being used?
Answer: Markup Language.

5: A developer wants to structure a webpage.
Which language is required?
Answer: HTML.

IMPORTANT POINTS

✓ HTML stands for HyperText Markup Language.
✓ HTML was developed by Tim Berners-Lee.
✓ HTML is a markup language.
✓ HTML uses tags.
✓ HTML provides webpage structure.
✓ HTML is not a programming language.
✓ HTML works with browsers.

SUMMARY

- HTML stands for HyperText Markup Language.
- HTML is used to create web pages.
- HTML uses tags and elements.
- HTML provides structure to web pages.
- HTML is not a programming language.
- HTML forms the foundation of web development.

HTML is the foundation of every web page, providing structure and organization to information displayed on the World Wide Web.

5. HTML ELEMENTS

Every webpage available on the World Wide Web is made up of different types of content such as headings, paragraphs, images, hyperlinks, tables, forms, and multimedia. Although these contents appear differently on the screen, a web browser cannot identify their purpose unless they are represented using HTML elements. HTML elements provide the structure and meaning required for the browser to interpret and display webpage content correctly. For this reason, HTML elements are considered the fundamental building blocks of every webpage.

When a user opens a college website, the browser automatically displays the college name as a heading, admission details as paragraphs, navigation options as hyperlinks, and images in appropriate locations. This is possible because each piece of information is enclosed within a suitable HTML element.

5.2 Need for HTML Elements

A browser processes only instructions written in HTML. If a webpage contains only plain text, the browser cannot distinguish between a heading, a paragraph, or a hyperlink. HTML elements solve this problem by defining the purpose of each content item. They organize the webpage into meaningful sections, improve readability, support accessibility, and help search engines understand webpage content.

For example, consider a college admission webpage. The college name should appear larger than the remaining text, admission details should be displayed as paragraphs, and the online application option should be presented as a clickable link. HTML elements enable the browser to display all these contents appropriately.

Example:
A university is designing its admission portal. The homepage must display the university name, admission notification, and an "Apply Online" link. Without HTML elements, the browser cannot identify which content is the title and which content should function as a hyperlink. By using appropriate HTML elements, the webpage becomes structured, readable, and user-friendly.

5.3 Definition of HTML Element

An HTML element is a complete unit of an HTML document that consists of an opening tag, content, and a closing tag, representing a specific part of a webpage.

Syntax

```
<tagname>Content</tagname>
```

Example

```
<p>Welcome Students</p>
```

In the above example,
Opening Tag:
Content: Welcome Students
Closing Tag:

The complete structure Welcome Students is called an **HTML element.**
It is important to understand that the browser reads the complete element rather than the content alone. The tags specify the type of content, while the enclosed text represents the information displayed to the user.

5.4 Structure of an HTML Element

A standard HTML element consists of three parts:
Opening Tag: Indicates the beginning of an HTML element.
Content: Represents the information that will be displayed on the webpage.
Closing Tag: Indicates the end of the HTML element.

Example
<h1>ABC College of Engineering</h1>

Structure
Opening Tag : <h1>
Content : ABC College of Engineering
Closing Tag : </h1>

When the browser encounters this element, it recognizes that the enclosed text is the main heading of the webpage and displays it accordingly.

Example
A college website needs to display its name as the main heading. The developer uses:
<h1>ABC College of Engineering</h1>

The browser interprets <h1> as a heading element and displays the college name in a larger font, making it immediately visible to users.

5.5 HTML Tags and HTML Elements

Students often confuse HTML tags with HTML elements. Although these terms are related, they are not identical. An **HTML tag** is the instruction enclosed within angle brackets (< >). It indicates the beginning or end of an HTML element. An **HTML element** is the complete structure formed by the opening tag, the content, and the closing tag.

Example

<h2>Department of Computer Science</h2>

Opening Tag : <h2>

Content : Department of Computer Science

Closing Tag : </h2>

Complete HTML Element :

<h2>Department of Computer Science</h2>

Thus, tags are components of an HTML element, while the complete combination forms the element itself.

5.6 Types of HTML Elements

HTML elements are commonly classified into the following categories:

5.6.1 Container Elements

Container elements contain information between opening and closing tags.

Example 1:

Welcome Students

Since the paragraph contains text enclosed between and , it is called a **container element**.

Example 2:

Department Information

Example

A hospital website displays patient instructions using paragraph elements because the instructions contain descriptive text enclosed within opening and closing tags.

5.6.2 Empty Elements

Empty elements do not contain any content. They perform a specific function without requiring a closing tag.

Examples

Creates a line break.

<hr>

Creates a horizontal line.

<img src="college.jpg">

Displays an image.

Example

A college website inserts a line break between the college name and address using the
 element.

5.6.3 Nested Elements

When one HTML element is placed inside another HTML element, it is called a **nested element.**

Example

```
<body>

<h1>ABC College</h1>

<p>Admissions Open</p>

</body>
```

Here, the <h1> and <p> elements are nested inside the <body> element.
Nested elements help organize webpage content logically.

5.6.4 Block-Level Elements

Block-level elements always begin on a new line and occupy the available horizontal width.

Example

```
<h1>ABC College</h1>
<p>Department of Artificial Intelligence</p>
```

The heading and paragraph appear on separate lines because both are block-level elements.

5.6.5 Inline Elements

Inline elements occupy only the required space and continue on the same line without creating a new line.

Example

```
<p>Welcome to <span>HTML</span> Programming.</p>
```

The element highlights only the word **HTML** while remaining within the same line.

Example

An online shopping website displays: "Price: ₹999" Only the price is highlighted using the element without affecting the surrounding text.

5.7 HTML Program

The following program demonstrates the use of different HTML elements.

```
<!DOCTYPE html>
<html>

<head>
      <title>HTML Elements</title>
</head>

<body>

<h1>ABC College of Engineering</h1>

<p>Admissions are open for the academic year 2026.</p>

<p>Visit the <a href="admission.html">Admission Portal</a> for online registration.</p>

<hr>

<p>Contact Us<br>Phone: 9876543210</p>

</body>

</html>
```

Program Explanation

The <h1> element displays the main heading. The elements display descriptive information. The <a> element creates a hyperlink for online admission. The <hr> element inserts a horizontal line, while
 moves the phone number to the next line. Together, these elements create a simple and well-structured webpage.

5.8 Scenario-Based Questions

Scenario 1

A college wants to display its name as the largest text on the homepage.

Question: Which HTML element should be used?

Answer: <h1>ABC College</h1>

Scenario 2

A hospital website wants to display patient instructions in sentence form.

Question: Which HTML element is suitable?

Answer: Take the prescribed medicine twice a day.

Scenario 3

A shopping website wants to highlight only the product price without affecting the remaining text.

Question: Which HTML element is appropriate?

Answer: ₹999

Scenario 4

A webpage needs to display the company logo.

Question: Which HTML element should be used?

Answer: <img src="logo.png">

5.9 Important Questions

Question 1: What is an HTML element?

Answer: An HTML element is a complete unit of an HTML document consisting of an opening tag, content, and a closing tag.

Question 2: Differentiate between an HTML tag and an HTML element.

Answer: An HTML tag is an instruction enclosed within angle brackets, whereas an HTML element is the complete structure consisting of the opening tag, content, and closing tag.

Question 3: What is a nested element?

Answer: A nested element is an HTML element placed inside another HTML element.

5.10 Summary

HTML elements provide the basic structure of every webpage. They help browsers identify and display different types of content such as headings, paragraphs, hyperlinks, and images. A standard HTML element consists of an opening tag, content, and a closing tag. Depending on their purpose, HTML elements are classified as container elements, empty elements, nested elements, block-level elements, and inline elements. A clear understanding of HTML elements forms the foundation for learning HTML tags, attributes, and webpage design.

6. HTML Tags and Attributes

6.1 Introduction

HTML elements define the structure of a webpage, but they are created using **HTML tags**. Tags identify the beginning and end of an element, while **attributes** provide additional information about that element. Together, tags and attributes help browsers display webpage content correctly and enable developers to control the behaviour and appearance of HTML elements.

For example, when an image is displayed on a webpage, the browser must know where the image is stored. This information is provided through an attribute within the <img> tag.

6.2 HTML Tags

Definition: An HTML tag is a predefined keyword enclosed within angle brackets (< >) that instructs the web browser how to interpret and display webpage content.

HTML tags are the basic commands of HTML. They identify different webpage components such as headings, paragraphs, tables, forms, images, and hyperlinks. Most HTML tags are written in pairs, consisting of an opening tag and a closing tag.

Syntax

```
<tagname>Content</tagname>
```

Example

```
<h1>Welcome Students</h1>
```

In the above example,

Opening Tag : <h1>
Content : Welcome Students
Closing Tag : </h1>

The browser reads these tags and displays the enclosed content as a heading.

6.3 Types of HTML Tags

HTML tags are broadly classified into two categories.

6.3.1 Paired Tags

Paired tags consist of an opening tag and a corresponding closing tag. The content is enclosed between these two tags.

Example

HTML is a Markup Language.
Other examples include:
<h1>Heading</h1>

<title>Web Page</title>

Content

Scenario-Based Example

A college website displays the department name as a heading. Since the heading contains text, the developer uses paired tags.

<h2>Department of Computer Science</h2>

6.3.2 Empty (Unpaired) Tags

Empty tags do not contain any content and therefore do not require a closing tag. They perform a specific function immediately.

Examples

<hr>

<img src="college.jpg">

The
 tag inserts a line break.
The <hr> tag displays a horizontal line.
The <img> tag displays an image.

Scenario-Based Example

A hospital website displays the hospital logo at the top of every page. The developer uses the <img> tag because it displays an image without enclosing any content.

6.4 HTML Attributes

Definition: An HTML attribute is a property that provides additional information about an HTML element and modifies its behaviour or appearance.

Attributes are always written inside the opening tag and usually consist of a **name** and a **value**.

Syntax

<tagname attribute="value">Content</tagname>

Example

Visit College Website

Here,

Attribute Name : href

Attribute Value : https://www.xyzcollege.edu

The href attribute tells the browser the destination of the hyperlink.

6.5 Common HTML Attributes

Some commonly used HTML attributes are: **href**

Specifies the destination address of a hyperlink.

Example

Apply Online

Src - Specifies the location of an image, audio, or video file.

Example 1:

<img src="logo.png">

alt

Provides alternative text if an image cannot be displayed.

Example 2:

<img src="college.jpg" alt="College Building">

id

Assigns a unique identifier to an HTML element.

Example 3:

Welcome Students

class

Groups multiple HTML elements for applying common styles.

Example 4:

Admissions Open

6.6 HTML Program Using Tags and Attributes

<!DOCTYPE html>

```
<html>

<head>
      <title>HTML Tags and Attributes</title>
</head>

<body>

<h1>ABC College of Engineering</h1>

<p>Admissions are open for 2026.</p>

<img src="college.jpg" alt="College Campus">

<br>

<a href="admission.html">Apply Online</a>

</body>

</html>
```

Program Explanation

The <h1> tag displays the college name as the main heading. The <p> tag displays admission information. The <img> tag displays the college image using the src attribute, while the alt attribute provides alternative text. The
 tag inserts a line break, and the <a> tag creates a hyperlink using the href attribute.

6.7 Scenario-Based Questions

Scenario 1

A university website wants students to move from the homepage to the online admission page.

Question: Which HTML tag and attribute should be used?

Answer:

```
<a href="admission.html">Apply Online</a>
```

Scenario 2

A shopping website wants to display a product image.

Question: Which HTML tag and attribute are required?

Answer:

```
<img src="mobile.jpg" alt="Mobile Phone">
```

Scenario 3

A webpage needs to move the next sentence to a new line.

Question: Which HTML tag should be used?

Answer:

Scenario 4

A company website wants to identify a paragraph uniquely for styling or scripting.

Question: Which attribute should be used?

Answer:

Welcome to ABC Pvt. Ltd.

6.8 Important Questions

Question 1: What is an HTML tag?

Answer: An HTML tag is a predefined keyword enclosed within angle brackets that instructs the browser how to display webpage content.

Question 2: What is an HTML attribute?

Answer: An HTML attribute is a property that provides additional information about an HTML element.

Question 3: Differentiate between an HTML tag and an HTML attribute.

Answer: An HTML tag defines an HTML element, whereas an HTML attribute provides additional information or modifies the behaviour of that element.

Question 4: Name any four commonly used HTML attributes.

Answer: href, src, alt, and id.

6.9 Summary

HTML tags are the fundamental instructions used to create webpage elements, while HTML attributes provide additional information that controls the behaviour or appearance of those elements. Understanding tags and attributes is essential for developing structured, interactive, and user-friendly web pages. In the next topic, we will learn how HTML headings and paragraphs help organize textual content on a webpage.

7. HTML Headings and Paragraphs

7.1 Introduction

Every webpage contains textual information that must be presented in a clear and organized manner. Whether it is a college website, an online shopping portal, or a news website, information is usually divided into titles, headings, subheadings, and paragraphs. HTML provides specific elements to organize text so that users can easily understand the content. The <h1> to <h6> elements are used to create headings of different importance, while the <p> element is used to display paragraphs. Proper use of headings and paragraphs improves readability, accessibility, and search engine optimization (SEO).

7.2 HTML Headings

Definition: HTML headings are predefined elements used to define titles and subheadings on a webpage. They help organize content into different levels of importance.

HTML provides six heading levels, from <h1> to <h6>. The <h1> element represents the highest and most important heading, whereas <h6> represents the lowest heading level.

Syntax

```
<h1>Main Heading</h1>
```

Example

```
<h1>ABC College of Engineering</h1>

<h2>Department of Artificial Intelligence</h2>

<h3>Faculty Information</h3>
```

The browser displays <h1> as the largest heading and gradually decreases the size and importance from <h2> to <h6>.

Why are Heading Levels Important?

Headings organize webpage content into logical sections. They help users quickly identify the main topic and its subtopics. Search engines also use headings to understand the structure of a webpage, making proper heading usage important for SEO.

Example

A university website displays the university name, department name, and laboratory details.

```
<h1>ABC University</h1>

<h2>Department of Computer Science</h2>

<h3>Artificial Intelligence Laboratory</h3>
```

Here, the browser understands the hierarchy of information and displays it accordingly.

7.3 HTML Paragraphs

Definition: An HTML paragraph is a block-level element used to display textual information in the form of sentences or paragraphs on a webpage.

Paragraphs are created using the element. Each paragraph begins on a new line and automatically leaves a small amount of space before and after it.

Syntax

Paragraph Content

Example

ABC College of Engineering offers undergraduate and postgraduate programs in various disciplines. . The browser displays the enclosed text as a paragraph with proper spacing.

Importance of Paragraphs

Paragraphs improve the readability of webpages by presenting information in a structured manner. Long blocks of continuous text become easier to read when divided into paragraphs.

Example

A hospital website provides health guidelines to patients. Instead of displaying all information as a single block of text, each guideline is written as a separate paragraph, making it easier for patients to read and understand.

7.4 Difference Between Headings and Paragraphs

Headings are used to introduce a topic or section, whereas paragraphs provide detailed information about that topic.

Example,

<h1>Artificial Intelligence</h1>

Artificial Intelligence is a branch of computer science that enables machines to perform tasks requiring human intelligence.

7.5 HTML Program Using Headings and Paragraphs

<!DOCTYPE html>

<html>

<head>

<title>Headings and Paragraphs</title>

</head>

```
<body>

<h1>ABC College of Engineering</h1>

<h2>Department of Computer Science</h2>

<p>Admissions are open for the academic year 2026. Students can apply through the online admission portal.</p>

<h2>Courses Offered</h2>

<p>The department offers undergraduate, postgraduate, and research programmes in Computer Science.</p>

</body>

</html>
```

Program Explanation

The <h1> element displays the college name as the main heading. The <h2> elements divide the webpage into meaningful sections. The <p> elements provide detailed information related to each heading. This combination creates a well-structured and readable webpage.

7.6 Scenario-Based Questions

Scenario 1

A college website wants to display the college name as the most important title.

Question: Which HTML element should be used?

Answer:

```
<h1>ABC College of Engineering</h1>
```

Scenario 2

A news website displays the article title followed by the news content.

Question: Which HTML elements are suitable?

Answer:

```
<h1>Heavy Rainfall in Bengaluru</h1>

<p>Several parts of the city experienced heavy rainfall throughout the day.</p>
```

Scenario 3

An educational website displays chapter titles followed by detailed explanations.

Question: Which HTML elements should be used?

Answer:
Headings should be created using <h1> to <h6>, and explanations should be written using the <p> element.

Scenario 4
A company website wants to organize information into different sections such as "About Us", "Services", and "Contact".
Question: Which HTML element is suitable for displaying these section titles?
Answer:
Heading elements (<h1> to <h6>) should be used depending on the importance of each section.

7.7 Important Questions

Question 1: What are HTML headings?
Answer: HTML headings are elements used to define titles and subheadings on a webpage. They range from <h1> to <h6>.

Question 2: Which HTML heading represents the highest level?
Answer: <h1> represents the highest and most important heading.

Question 3: What is an HTML paragraph?
Answer: An HTML paragraph is a block-level element used to display textual information in sentence or paragraph form using the <p> element.

Question 4: Why are headings important?
Answer: Headings organize webpage content, improve readability, and help search engines understand the structure of the webpage.

Question 5: Which HTML element is used to display normal textual information?
Answer: The <p> element.

7.8 Summary

HTML headings and paragraphs are essential elements for organizing textual content on a webpage. Headings define the structure and hierarchy of information, while paragraphs provide detailed explanations. Proper use of these elements improves readability, accessibility, and search engine optimization. In the next topic, we will learn how **HTML Hyperlinks and Lists** help users navigate webpages and organize information efficiently.

8. HTML Hyperlinks and Lists

8.1 Introduction

Modern websites consist of multiple webpages connected together. For example, a college website contains separate pages for Home, About Us, Departments, Admissions, Faculty, and Contact Us. A user can move from one page to another simply by clicking a link. Similarly, information such as course names, product features, or instructions is often displayed in an organized manner using lists. HTML provides **hyperlinks** for navigation and **lists** for presenting related information in a structured format. These elements improve user experience and make webpages easier to understand.

8.2 HTML Hyperlinks

Definition: A hyperlink is an HTML element that connects one webpage to another webpage, document, image, email address, or external website.

Hyperlinks are created using the <a> (anchor) element. When a user clicks a hyperlink, the browser opens the specified destination. The destination is defined using the **href** attribute.

Syntax

```
<a href="URL">Link Text</a>
```

Example

```
<a href="admission.html">Apply Online</a>
```

In the above example,
<a> creates the hyperlink.
href specifies the destination page.
"Apply Online" is the clickable text displayed on the webpage.

Example

A college website provides an "Apply Online" option on its homepage. When students click the link, they should be redirected to the admission page.

```
<a href="admission.html">Apply Online</a>
```

The browser opens **admission.html** when the user clicks the link.

8.3 Types of Hyperlinks

Hyperlinks can connect different types of resources depending on the requirement.

Internal Hyperlink

An internal hyperlink connects one webpage to another webpage within the same website.

Example

```
<a href="about.html">About Us</a>
```

Example

A university website contains separate pages for "Home", "Departments", and "Contact". Clicking "Departments" opens another page within the same website.

External Hyperlink

An external hyperlink connects to another website.

Example

```
<a href="https://www.w3.org">W3C Website</a>
```

Example

A college website provides a link to the AICTE official website for students seeking additional information.

Email Hyperlink

An email hyperlink opens the user's default email application.

Example

```
<a href="mailto:info@abccollege.edu">Email Us</a>
```

Example

A student wants to contact the admission office by clicking an email link instead of typing the email address manually.

8.4 HTML Lists

Definition: An HTML list is a collection of related items displayed in a structured format. Lists make information easier to read and understand. They are commonly used to display menus, product features, instructions, departments, and services.

HTML provides three types of lists:

- Ordered List
- Unordered List
- Description List

8.5 Ordered List

Definition: An ordered list displays items in a specific sequence using numbers or letters. It is created using the <ol> element, and each list item is defined using the <li> element.

Syntax

```
<ol>

<li>Item 1</li>

<li>Item 2</li>

</ol>
```

Example

```
<ol>

<li>Registration</li>

<li>Document Verification</li>

<li>Fee Payment</li>

</ol>
```

The browser displays:

1. Registration
2. Document Verification
3. Fee Payment

Example

A college admission portal displays the admission process step by step. Since the sequence is important, an ordered list is used.

8.6 Unordered List

Definition: An unordered list displays items using bullets when the order of items is not important. It is created using the <ul> element.

Syntax

```
<ul>

<li>Item 1</li>

<li>Item 2</li>

</ul>
```

Example

```
<ul>

<li>Computer Science</li>

<li>Artificial Intelligence</li>

<li>Data Science</li>
```


The browser displays bullet points for each department.

Example

A university website lists the departments offered by the institution. Since the departments have no specific order, an unordered list is suitable.

8.7 Description List

Definition: A description list displays a term along with its corresponding description. It uses the <dl>, <dt>, and <dd> elements.

Syntax

```
<dl>

<dt>Term</dt>

<dd>Description</dd>

</dl>
```

Example

```
<dl>

<dt>HTML</dt>

<dd>HyperText Markup Language</dd>

<dt>WWW</dt>

<dd>World Wide Web</dd>

</dl>
```

Example

An educational website explains technical terms and their meanings. A description list helps present the information clearly.

8.8 HTML Program

```
<!DOCTYPE html>

<html>

<head>
```

```
<title>Hyperlinks and Lists</title>

</head>

<body>

<h1>ABC College of Engineering</h1>

<p>Quick Links</p>

<a href="admission.html">Apply Online</a>

<h2>Departments</h2>

<ul>

<li>Computer Science</li>

<li>Artificial Intelligence</li>

<li>Information Technology</li>

</ul>

<h2>Admission Process</h2>

<ol>

<li>Register Online</li>

<li>Upload Documents</li>

<li>Pay Fees</li>

</ol>

</body>

</html>
```

Explanation

The <a> element creates a hyperlink to the admission page. The <ul> element displays the list of departments using bullets, while the <ol> element presents the admission procedure

in a numbered sequence. Together, these elements make the webpage easy to navigate and understand.

8.9 Scenario-Based Questions

Scenario 1
A college website wants students to move from the homepage to the online admission page.
Question: Which HTML element should be used?
Answer:
Apply Online

Scenario 2
A shopping website wants to display product features as bullet points.
Question: Which HTML list is suitable?
Answer:
An unordered list (<ul>).

Scenario 3
A university website displays the admission procedure in the correct sequence.
Question: Which HTML list is appropriate?
Answer:
An ordered list (<ol>).

Scenario 4
An educational website explains programming terms with their meanings.
Question: Which HTML list should be used?
Answer:
A description list (<dl>).

8.10 Important Questions with Answers

Question 1: What is a hyperlink?
Answer: A hyperlink is an HTML element that connects one webpage to another webpage, document, or website.

Question 2: Which HTML element is used to create hyperlinks?
Answer: The <a> (anchor) element.

Question 3: What is the purpose of the href attribute?
Answer: It specifies the destination address of a hyperlink.

Question 4: Name the three types of HTML lists.
Answer: Ordered List, Unordered List, and Description List.

Question 5: Which HTML element is used to create an ordered list?

Answer: The <ol> element.

8.11 Summary

Hyperlinks and lists are essential HTML elements that improve webpage navigation and content organization. Hyperlinks allow users to move between webpages and websites, while lists present related information in a clear and structured manner. Depending on the requirement, HTML provides ordered lists, unordered lists, and description lists. Proper use of hyperlinks and lists makes webpages more user-friendly, organized, and easier to navigate. In the next topic, we will learn how **HTML Tables** are used to display information in rows and columns.

9. HTML Tables

9.1 Introduction

In many web applications, information needs to be presented in a systematic and organized manner. For example, a college website displays student details, a hospital website maintains patient records, and an e-commerce website lists product information. Displaying such information as plain text makes it difficult to read and compare. HTML solves this problem by providing **tables**, which organize data into rows and columns. Tables improve readability and help users quickly locate and compare information.

9.2 HTML Tables

Definition: An HTML table is a collection of related data arranged in rows and columns using HTML table elements.

A table consists of rows and columns. Each row contains one or more cells that store related information. HTML tables are created using the <table> element, while rows and cells are represented using <tr>, <th>, and <td> elements.

9.3 Structure of an HTML Table

An HTML table is created using different elements, each having a specific purpose.

<table> - Defines the complete table.

<tr> (Table Row) - Represents a row in the table.

<th> (Table Heading) - Represents the heading of a column. By default, browsers display table headings in bold and center alignment.

<td> (Table Data) - Represents the actual data stored in each cell of the table.

Example

```
<table>

<tr>

<th>Student Name</th>

<th>Course</th>

</tr>

<tr>
```

<td>Rahul</td>

<td>B.Tech AI</td>

</tr>

</table>
In the above example,

- <table> creates the table.
- <tr> creates two rows.
- <th> creates the column headings.
- <td> stores the student information.

9.4 Why Are HTML Tables Used?

Tables are used whenever related information needs to be displayed in a structured format. They improve readability and make comparison easier because information is arranged into rows and columns.

For example, a university website may display student attendance, examination results, or faculty details using tables. Similarly, online shopping websites use tables to compare product specifications, prices, and availability.

Example
A college wants to display faculty information on its website.

Instead of writing:
Ravi – Professor
Anita – Associate Professor
Kiran – Assistant Professor
the information can be displayed in a table.

<table>

<tr>

<th>Faculty Name</th>

<th>Designation</th>

</tr>

<tr>

```
<td>Ravi</td>

<td>Professor</td>

</tr>

<tr>

<td>Anita</td>

<td>Associate Professor</td>

</tr>

<tr>

<td>Kiran</td>

<td>Assistant Professor</td>

</tr>

</table>
```

The table presents the information in a much clearer and organized manner.

9.5 Table Border

By default, HTML tables may not display borders in older examples commonly found in textbooks. A border helps distinguish rows and columns, making the table easier to read.

Example

```
<table border="1">

<tr>

<th>Course</th>

<th>Duration</th>

</tr>

<tr>

<td>B.Tech</td>
```

```
<td>4 Years</td>

</tr>

</table>
```

The browser displays a table with visible borders around each cell.

Note: In modern HTML5 development, borders are usually applied using CSS instead of the border attribute. However, the border attribute is still commonly used in introductory HTML examples to help beginners understand table structure.

9.6 HTML Program

```
<!DOCTYPE html>

<html>

<head>

<title>Student Information</title>

</head>

<body>

<h1>ABC College of Engineering</h1>

<table border="1">

<tr>

<th>Student Name</th>

<th>Department</th>

<th>Year</th>

</tr>

<tr>

<td>Rahul</td>
```

```
<td>Artificial Intelligence</td>

<td>II</td>

</tr>

<tr>

<td>Priya</td>

<td>Computer Science</td>

<td>I</td>

</tr>

<tr>

<td>Karan</td>

<td>Data Science</td>

<td>III</td>

</tr>

</table>

</body>

</html>
```

Explanation

The <table> element creates the table structure. The first row contains <th> elements that display the column headings. The remaining rows use <td> elements to display student information. The border="1" attribute makes the table borders visible, helping users read the information easily.

9.7 Scenario-Based Questions

Scenario 1

A hospital website wants to display patient details such as Patient ID, Name, and Doctor in an organized format.

Question: Which HTML element should be used?

Answer:

The <table> element.

Scenario 2

A university website wants to display the headings "Student Name", "Department", and "CGPA".

Question: Which HTML element should be used for the column headings?

Answer:

The <th> element.

Scenario 3

An online shopping website wants to display product details such as Product Name, Price, and Stock Availability.

Question: Which HTML element stores the actual product information?

Answer:

The <td> element.

Scenario 4

A school website wants to create a new row for each student's information.

Question: Which HTML element should be used?

Answer:

The <tr> element.

9.8 Important Questions

Question 1: What is an HTML table?

Answer: An HTML table is a collection of related data arranged in rows and columns using HTML table elements.

Question 2: Which HTML element creates a table?

Answer: <table>

Question 3: What is the purpose of the <tr> element?

Answer: It defines a row in an HTML table.

Question 4: What is the difference between <th> and <td>?

Answer: <th> defines a table heading, whereas <td> stores the actual table data.

Question 5: Why are HTML tables used?

Answer: HTML tables are used to organize related information into rows and columns, making the data easier to read, compare, and understand.

9.9 Summary

HTML tables provide an effective way to organize and present related information in rows and columns. They are widely used to display student records, employee details, product information, timetables, and reports. The <table>, <tr>, <th>, and <td> elements work together to create a structured table that improves readability and data presentation. In the next topic, we will learn how **HTML Forms and Form Controls** are used to collect information from users through webpages.

10. HTML Forms and Form Controls

10.1 Introduction

Many websites not only display information but also allow users to enter information. For example, students fill out online admission forms, customers register on shopping websites, patients book hospital appointments, and users log in to online applications. Such interactions are possible through **HTML forms**. A form acts as a bridge between the user and the website by collecting information through various input controls. HTML provides different form controls to accept text, passwords, selections, and other user inputs in a structured manner.

10.2 HTML Forms

Definition: An HTML form is a section of a webpage used to collect information from users through various input controls.

The information entered in a form can be sent to another webpage for further processing. A form is created using the <form> element, which acts as a container for different form controls.

Syntax

<form>

Form Controls

</form>

Example

<form>

Student Name:
<input type="text">

</form>

In this example, the <form> element contains a text box that allows the user to enter a student name.

10.3 Form Attributes

The <form> element uses attributes to define how the form behaves.

action Attribute

Definition: The action attribute specifies the webpage that will receive the form data after the user clicks the Submit button.

Example

```
<form action="admission.html">
```

In this example, after the user submits the form, the browser opens the **admission.html** page.

Scenario-Based Example

A college admission form collects student details. After clicking the **Submit** button, the browser opens the **admission.html** page where the next step of the admission process begins.

method Attribute

Definition: The method attribute specifies how the browser sends the form data.
The two commonly used methods are **GET** and **POST**.

- **GET** sends the entered information through the webpage address (URL). It is generally used for non-sensitive information such as search keywords.
- **POST** sends the entered information securely in the request body. It is commonly used for confidential information such as passwords and personal details.

Example

```
<form action="admission.html" method="post">
```

In this example, the browser sends the entered information using the **POST** method and then opens the **admission.html** page.

Note: At this stage, students only need to understand the purpose of the method attribute. Processing the submitted information using server-side technologies will be learned in advanced web development.

10.4 HTML Form Controls

HTML provides different form controls to collect different types of user information.

1. Text Box

Definition: A text box is a form control used to accept single-line text input from the user.

Example

```
<input type="text">
```

Example

A university admission form asks students to enter their full name. Since the information is entered in a single line, a text box is used.

2. Password Field

Definition: A password field is a form control used to accept confidential information by hiding the entered characters.

Example

```
<input type="password">
```

Scenario-Based Example

A student logs into the college portal by entering a password. The characters are hidden to maintain privacy.

3. Radio Button

Definition: A radio button is a form control that allows the user to select only one option from a group of options.

Example

```
<input type="radio" name="gender"> Male
<input type="radio" name="gender"> Female
```

Scenario-Based Example

A student registration form asks applicants to select their gender. Since only one option can be chosen, radio buttons are used.

4. Checkbox

Definition: A checkbox is a form control that allows the user to select one or more options.

Example

```
<input type="checkbox"> Cricket
<input type="checkbox"> Football
<input type="checkbox"> Music
```

Scenario-Based Example

A college club registration form allows students to join multiple clubs. Therefore, checkboxes are suitable.

5. Drop-down List

Definition: A drop-down list is a form control that allows the user to select one option from a predefined list.

Example

```
<select>

<option>B.Tech AI</option>
```

```
<option>B.Tech CSE</option>

<option>B.Tech IT</option>

</select>
```

Scenario-Based Example

A college admission portal allows students to choose their preferred course from the available programmes.

6. Submit Button

Definition: A submit button is used to send the entered form information to the webpage specified in the action attribute.

Example

```
<input type="submit" value="Submit">
```

7. Reset Button

Definition: A reset button clears all the information entered by the user in the form.

Example

```
<input type="reset" value="Reset">
```

10.5 HTML Program

```
<!DOCTYPE html>
<html>

<head>
    <title>Student Admission Form</title>
</head>

<body>

<h1>ABC College Admission Form</h1>

<form action="admission.html">

Student Name:
<input type="text"><br><br>

Password:
<input type="password"><br><br>
```

```
Gender:
<input type="radio" name="gender"> Male
<input type="radio" name="gender"> Female

<br><br>

Course:
<select>
<option>B.Tech AI</option>
<option>B.Tech CSE</option>
<option>B.Tech IT</option>
</select>

<br><br>

Hobbies:
<input type="checkbox"> Sports
<input type="checkbox"> Music

<br><br>

<input type="submit" value="Submit">

<input type="reset" value="Reset">

</form>

</body>

</html>
```

Program Explanation

The <form> element creates the admission form. A text box is used to collect the student's name, while the password field accepts confidential information. Radio buttons allow the selection of one gender, the drop-down list helps the student choose a course, and checkboxes allow multiple hobbies to be selected. Finally, the **Submit** button sends the form to **admission.html**, and the **Reset** button clears all the entered information.

10.6 Scenario-Based Questions

Scenario 1

A college admission portal asks students to enter their full name.

Question: Which HTML form control should be used?

Answer:

```
<input type="text">
```

Scenario 2

A banking website asks users to enter their password.

Question: Which HTML form control is suitable?

Answer:

<input type="password">

Scenario 3

An online application form asks students to select their branch from a list of available courses.

Question: Which HTML form control should be used?

Answer:

A **drop-down list** (<select>).

Scenario 4

A sports club registration form allows students to select multiple games.

Question: Which HTML form control is appropriate?

Answer:

Checkbox (<input type="checkbox">).

Scenario 5

A student registration form asks applicants to select either **Male** or **Female**.

Question: Which HTML form control should be used?

Answer:

Radio button (<input type="radio">).

10.7 Important Questions

Question 1: What is an HTML form?

Answer: An HTML form is a section of a webpage used to collect information from users through various input controls.

Question 2: Which HTML element is used to create a form?

Answer: <form>

Question 3: What is the purpose of the action attribute?

Answer: It specifies the webpage that receives the form data after the user clicks the Submit button.

Question 4: Differentiate between a radio button and a checkbox.

Answer: A radio button allows the selection of only one option, whereas a checkbox allows the selection of one or more options.

Question 5: Which form control is used to select one option from a predefined list?
Answer: The **drop-down list** (<select>).

10.8 Summary

HTML forms are used to collect information from users through webpages. By combining the <form> element with form controls such as text boxes, password fields, radio buttons, checkboxes, drop-down lists, and buttons, developers can create interactive web pages that accept user input. Forms are widely used in online admissions, registrations, login pages, surveys, and booking applications. A clear understanding of HTML forms and form controls is essential before learning advanced web technologies.

11. Semantic HTML Elements

11.1 Introduction

As websites became larger and more complex, developers needed a better way to organize webpage content. Earlier, most webpages were designed using generic elements such as , making it difficult to understand the purpose of different sections. HTML5 introduced **Semantic HTML elements**, which clearly describe the role of each section of a webpage. Semantic elements improve webpage organization, make the code easier to read, enhance accessibility, and help search engines understand webpage content more effectively.

11.2 Semantic HTML Elements

Definition: Semantic HTML elements are HTML elements that clearly describe the meaning and purpose of the content they contain.

Unlike generic elements, semantic elements indicate the role of each section of a webpage. For example, a <header> element represents the header section, while a <footer> element represents the footer section.

11.3 Why are Semantic HTML Elements Important?

In earlier versions of HTML, developers commonly used elements for every section of a webpage. Although this displayed the webpage correctly, it did not explain the purpose of each section.

Semantic HTML elements solve this problem by giving meaningful names to different parts of the webpage. This improves code readability, makes website maintenance easier, supports screen readers for visually impaired users, and helps search engines identify important content.

Scenario-Based Example

A college website contains the following sections:

- College Logo
- Navigation Menu
- Department Information
- Latest News
- Contact Information

If every section is created using only elements, another developer may find it difficult to understand the webpage structure. By using semantic elements such as <header>, <nav>, <section>, and <footer>, the purpose of each section becomes immediately clear.

11.4 Common Semantic HTML Elements

1. <header>

Definition: The <header> element represents the introductory section of a webpage or a section of a webpage.

It usually contains the website logo, title, or introductory information.

Example

```
<header>

<h1>ABC College of Engineering</h1>

</header>
```

Scenario-Based Example

A university website displays its logo and university name at the top of every webpage. This information is placed inside the <header> element.

2. <nav>

Definition: The <nav> element represents the navigation section containing links to other webpages.

Example

```
<nav>

<a href="home.html">Home</a>

<a href="about.html">About</a>

<a href="contact.html">Contact</a>

</nav>
```

Scenario-Based Example

A hospital website provides navigation links for Home, Doctors, Appointments, and Contact pages using the <nav> element.

3. <section>

Definition: The <section> element represents a thematic section of related content within a webpage.

Example

```
<section>

<h2>Department of Artificial Intelligence</h2>

<p>Admissions are open.</p>

</section>
```

Scenario-Based Example

A college website displays separate sections for each department. Each department's information is enclosed within a <section> element.

4. <article>

Definition: The <article> element represents independent, self-contained content that can be distributed or reused independently.

Example

```
<article>

<h2>Annual Sports Meet</h2>

<p>The annual sports meet will be conducted next month.</p>

</article>
```

Scenario-Based Example

A news portal publishes each news article using the <article> element because every article is independent.

5. <footer>

Definition: The <footer> element represents the bottom section of a webpage containing copyright, contact information, or other related details.

Example

```
<footer>

<p>© 2026 ABC College. All Rights Reserved.</p>

</footer>
```

Scenario-Based Example

A company website displays its copyright information and contact details at the bottom of every webpage using the <footer> element.

11.5 HTML Program

```
<!DOCTYPE html>

<html>

<head>

<title>Semantic HTML</title>

</head>

<body>

<header>

<h1>ABC College of Engineering</h1>

</header>

<nav>

<a href="home.html">Home</a>

<a href="courses.html">Courses</a>

<a href="contact.html">Contact</a>

</nav>

<section>

<h2>Department of Computer Science</h2>

<p>Admissions are open for the academic year 2026.</p>

</section>

<footer>

<p>© 2026 ABC College</p>

</footer>
```

```
</body>

</html>
```

Program Explanation

The <header> element displays the college title at the top of the webpage. The <nav> element provides navigation links to different pages. The <section> element contains the department information, while the <footer> element displays the copyright information at the bottom of the webpage. Together, these semantic elements create a well-organized and meaningful webpage structure.

11.6 Scenario-Based Questions

Scenario 1

A university website wants to display the university logo and title at the top of every webpage.

Question: Which semantic HTML element should be used?

Answer:

<header>

Scenario 2

A shopping website wants to provide links for Home, Products, Cart, and Contact.

Question: Which semantic HTML element is appropriate?

Answer:

<nav>

Scenario 3

A news website publishes independent news reports.

Question: Which semantic HTML element should be used?

Answer:

<article>

Scenario 4

A college website displays copyright information at the bottom of the webpage.

Question: Which semantic HTML element should be used?

Answer:

<footer>

11.7 Important Questions

Question 1: What are Semantic HTML elements?

Answer: Semantic HTML elements are HTML elements that clearly describe the meaning and purpose of the content they contain.

Question 2: Why are Semantic HTML elements important?
Answer: They improve webpage structure, readability, accessibility, and search engine optimization by providing meaningful content organization.

Question 3: Which semantic HTML element is used for navigation links?
Answer: <nav>

Question 4: Which semantic HTML element represents the bottom section of a webpage?
Answer: <footer>

Question 5: Which semantic HTML element is used to represent an independent article?
Answer: <article>

11.8 Summary

Semantic HTML elements provide meaningful structure to webpages by clearly identifying different sections such as the header, navigation menu, content area, articles, and footer. Unlike generic elements, semantic elements improve code readability, accessibility, website maintenance, and search engine optimization. Using semantic HTML is considered a best practice in modern web development because it creates well-structured, understandable, and user-friendly webpages. In the next topic, we will learn about ** and Tags**, which are commonly used as generic containers for grouping and styling webpage content.

12. and Tags

12.1 Introduction

While developing a webpage, it is often necessary to group related content together for better organization and styling. For example, a webpage may contain a navigation menu, a content area, a sidebar, and a footer. Similarly, within a paragraph, only a few words may need special formatting. HTML provides the ** ** and **** elements for this purpose. Although both elements are used to group content, they differ in the way they display information. The element groups larger sections of a webpage, whereas the element groups a small portion of text within a line.

12.2 The Tag

Definition: The (Division) tag is a block-level HTML element used to group related content into separate sections of a webpage.

The element does not provide any visual formatting by itself. It acts as a container that helps organize webpage content and is commonly used with CSS and JavaScript to style or manipulate webpage sections.

Syntax

```
<div>

Content

</div>
```

Example

```
<div>

<h2>Department of Computer Science</h2>

<p>Admissions are open for the academic year 2026.</p>

</div>
```

In this example, the heading and paragraph are grouped into a single section using the element.

Scenario-Based Example

A college website contains separate sections for **Departments**, **Facilities**, and **Placements**. Each section can be grouped using a separate element, making the webpage well-organized and easier to design.

12.3 The Tag

Definition: The tag is an inline HTML element used to group or identify a small portion of text within a line.

Unlike the element, the element does not begin on a new line. It is mainly used to apply different styles or identify specific words without affecting the surrounding text.

Syntax

Content

Example

Welcome to HTML Programming.

In this example, only the words **HTML Programming** are enclosed within the element, while the remaining text continues on the same line.

Scenario-Based Example

An online shopping website displays:

Price: ₹999

The developer wants to highlight only the price without changing the remaining text. The element is used to identify only the price so that it can be styled separately.

12.4 Difference Between and

Although both elements are used for grouping content, they serve different purposes.
The element is a **block-level** element. It always starts on a new line and generally groups larger sections of a webpage.

The element is an **inline** element. It remains within the same line and is used to group a small portion of text.

Example

```
<div>

<h2>Student Details</h2>

<p>Name: Rahul</p>

</div>

<p>Course: <span>B.Tech AI</span></p>
```

In this example, the element groups the complete student information, whereas the element identifies only the course name within the paragraph.

12.5 HTML Program

```
<!DOCTYPE html>

<html>

<head>

<title>Div and Span</title>

</head>

<body>

<div>

<h1>ABC College of Engineering</h1>

<p>Admissions are open for the academic year 2026.</p>

</div>

<p>Department: <span>Artificial Intelligence</span></p>

</body>

</html>
```

Program Explanation

The element groups the college heading and admission information into one section of the webpage. The element groups only the words **Artificial Intelligence** within the paragraph. Although both elements group content, creates a separate block, whereas affects only a part of the text without starting a new line.

12.6 Scenario-Based Questions

Scenario 1

A college website wants to group the **About Us**, **Departments**, and **Placements** sections separately. **Question:** Which HTML element should be used?
Answer:

Scenario 2

A shopping website wants to highlight only the product price within a sentence.
Question: Which HTML element is suitable?
Answer:

Scenario 3

A university webpage contains multiple sections, each requiring different CSS styles.
Question: Which HTML element should be used to group each section?
Answer: The element.

Scenario 4

A news website wants to highlight only the word **Breaking News** within a paragraph.
Question: Which HTML element should be used?
Answer: The element.

12.7 Important Questions

Question 1: What is the purpose of the tag?
Answer: The tag is used to group related content into separate sections of a webpage.

Question 2: What is the purpose of the tag?
Answer: The tag is used to group or identify a small portion of text within a line.

Question 3: What is the main difference between and ?
Answer: The element is a block-level element used for grouping larger sections of a webpage, whereas the element is an inline element used for grouping a small portion of text within a line.

Question 4: Which element starts on a new line?
Answer: The element.

Question 5: Which element remains on the same line?
Answer: The element.

12.8 Summary

The and elements are generic HTML containers used to organize webpage content. The element groups larger sections of a webpage and starts on a new line, while the element groups a small portion of text without affecting the line structure. These elements are widely used in modern web development to create well-organized webpages and apply different styles to specific sections of content. In the next topic, we will study **Multimedia Elements and HTML5**, where we will learn how HTML5 enables webpages to display audio, video, and other multimedia content without using external plugins.

13. Multimedia Elements and HTML5

13.1 Introduction

Modern websites are no longer limited to displaying only text and images. They also provide audio lectures, video tutorials, online meetings, animations, and interactive content. For example, educational websites stream video lectures, music applications play songs, and news websites publish video reports. Earlier, displaying multimedia on webpages required external plugins such as Adobe Flash. HTML5 simplified this process by introducing built-in multimedia elements that allow browsers to play audio and video directly without additional software.

13.2 Multimedia in HTML

Definition: Multimedia refers to the combination of different types of digital content, such as text, images, audio, video, and animations, presented together on a webpage to improve user interaction and learning.

Multimedia makes webpages more attractive, interactive, and informative. It is widely used in educational portals, entertainment websites, online learning platforms, social media applications, and business websites.

Scenario-Based Example

An online learning platform provides recorded video lectures and audio explanations for students. Instead of downloading separate media players, students can watch videos and listen to audio directly in their web browsers because of HTML5 multimedia support.

13.3 HTML5 Multimedia Elements

HTML5 introduced dedicated elements for embedding multimedia content into webpages. The most commonly used multimedia elements are <audio> and <video>.

13.4 The <audio> Element

Definition: The <audio> element is used to embed and play audio files directly within a webpage.

It allows users to play, pause, and control audio without using external software.

Syntax

```
<audio controls>
<source src="audio.mp3" type="audio/mpeg">
</audio>
```

Example

```
<audio controls>
<source src="welcome.mp3" type="audio/mpeg">
</audio>
```

In this example, the browser displays an audio player with controls such as Play, Pause, and Volume.

Scenario-Based Example

A college website provides recorded welcome messages from the Principal. Students can listen to the message directly using the <audio> element.

13.5 The <video> Element

Definition: The <video> element is used to embed and play video files directly within a webpage.

HTML5 allows users to watch videos without installing external plugins.

Syntax

```
<video width="400" controls>

<source src="video.mp4" type="video/mp4">

</video>
```

Example

```
<video width="400" controls>

<source src="campus.mp4" type="video/mp4">

</video>
```

In this example, the browser displays a video player with controls such as Play, Pause, Volume, and Full Screen.

Scenario-Based Example

A university uploads a campus tour video on its website. Visitors can watch the video directly on the webpage using the <video> element.

13.6 What is HTML5?

Definition: HTML5 is the fifth and the latest major version of HyperText Markup Language, developed to support modern web applications by providing improved structure, multimedia support, graphics, and better user interaction.

HTML5 was introduced to overcome the limitations of earlier HTML versions and to support the growing needs of modern websites. It enables developers to create responsive and interactive webpages without relying on external plugins.

13.7 Features of HTML5

HTML5 introduced several important features that simplified modern web development.

- Native support for audio and video.
- New semantic elements such as <header>, <nav>, <section>, <article>, and <footer>.
- Better support for multimedia without external plugins.
- Improved form controls.
- Better compatibility with modern web browsers.
- Support for mobile-friendly and responsive websites.

Scenario-Based Example

A university decides to redesign its website. The new website should display video lectures, online application forms, and a mobile-friendly interface. HTML5 provides all these features, making it the preferred choice for modern web development.

13.8 HTML Program

```
<!DOCTYPE html>

<html>

<head>

<title>HTML5 Multimedia</title>

</head>

<body>

<h1>ABC College of Engineering</h1>

<h2>Welcome Message</h2>

<audio controls>

<source src="welcome.mp3" type="audio/mpeg">

</audio>

<h2>Campus Tour</h2>

<video width="400" controls>

<source src="campus.mp4" type="video/mp4">
```

</video>

</body>

</html>

Program Explanation

The webpage displays the college title using the <h1> element. The <audio> element plays the Principal's welcome message, while the <video> element displays the campus tour video. The controls attribute enables the browser to display playback controls such as Play, Pause, and Volume. HTML5 allows both media files to play directly in the browser without requiring external plugins.

13.9 Scenario-Based Questions

Scenario 1

An educational website wants students to listen to recorded lectures online.

Question: Which HTML5 element should be used?

Answer: <audio>

Scenario 2

A university website wants to display a campus tour video.

Question: Which HTML5 element should be used?

Answer: <video>

Scenario 3

A company upgrades its website to support multimedia without using Adobe Flash.

Question: Which version of HTML provides native multimedia support?

Answer: HTML5

Scenario 4

A news website wants visitors to watch live video reports directly in the browser.

Question: Which HTML5 element is suitable?

Answer:

<video>

13.10 Important Questions

Question 1: What is multimedia?

Answer: Multimedia is the combination of different types of digital content such as text, images, audio, video, and animations presented together on a webpage.

Question 2: What is the purpose of the <audio> element?

Answer: The <audio> element is used to embed and play audio files directly within a webpage.

Question 3: What is the purpose of the <video> element?
Answer: The <video> element is used to embed and play video files directly within a webpage.

Question 4: What is HTML5?
Answer: HTML5 is the fifth and the latest major version of HyperText Markup Language that provides improved support for multimedia, semantic elements, modern web applications, and responsive websites.

Question 5: Name any four features of HTML5.
Answer:

- Native audio support
- Native video support
- Semantic HTML elements
- Improved form controls

13.11 Summary

HTML5 has transformed web development by introducing built-in multimedia support and several modern features that improve website functionality and user experience. The <audio> and <video> elements allow developers to embed media files directly into webpages without relying on external plugins. In addition, HTML5 provides semantic elements, improved forms, and better browser compatibility, making it the standard language for developing modern, interactive, and responsive websites.